Infrastructure planning

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Appraising infrastructure modifications and adaptive plans

Project Sheet 5 (Weeks 9 and 10)

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0 Purpose and scope of Notebook 5

Notebook 4 compared static, predetermined staged and adaptive plans under the same transient futures. It established when packages are triggered, constructed and opened; whether performance requirements are protected; and what is gained or lost physically by waiting. Notebook 5 adds the discounted life-cycle appraisal needed to decide whether those outcomes are worth their cost to society and downside risk metrics including robustness and regret.

Notebook 05_appraisal.ipynb evaluates the six MehrSpur plans (Baseline, Static 1, Static 2, Staged 1, Staged 2 and Adaptive) across 200 paired uncertain scenarios. The Adaptive plan is compared in particular with Staged 1, the currently planned staged implementation, and with Static 2. The notebook compares incremental net present value (NPV), benefit and cost components, downside risk, economic regret and protection of the planning requirements. It then examines the value and price of waiting, diagnoses vulnerability mechanisms, tests appraisal assumptions and supports a final recommendation.

The objective is to understand:

- how the Notebook 04 scenarios, requirements, trigger rules and implementation dates are handed over to the appraisal;
- how matched welfare, the Rule of a Half, annualisation and discounting convert transport outcomes into societal value;
- how capital expenditure, operating and maintenance costs, adaptive option premiums, construction emissions and residual value are represented;
- how NPV distributions, component breakdowns, downside losses and regret reveal different dimensions of economic performance;
- why economic value and protection of planning requirements must remain separate;
- when adaptive waiting avoids unnecessary investment, when it merely defers expenditure and when it sacrifices benefits or objective protection;
- which uncertain scenario inputs and appraisal assumptions can overturn the economic judgement;
- how the complete evidence should be used to improve the plan and justify a recommendation.

The appraisal is incremental: every plan is compared with the Baseline in the same scenario over the 40-year analysis horizon, with costs and benefits discounted at 2% per year in the MehrSpur example. As it is not a complete account of every social, ecological or distributional effect. Unpriced effects could therefore be recorded alongside the monetary results (e.g., as performance requirements).

0.1 How to use the Project Sheets

The present project sheet provides the conceptual guidance and required outputs for the fifth project notebook (Weeks 9–10). In Notebook 05, adapt the assumptions, monetary values, project costs, requirements and interpretations to your own case study.

Tasks are grouped into three categories, as usual. Mandatory tasks provide evidence that every group is expected to produce. Selected results should go in the final report. Supporting tasks build understanding and help check the analysis. Advanced tasks extend the supplied workflow where time, data and modelling confidence permit.

Supporting Task (optional) Designed to help you understand the appraisal method, become familiar with the notebook outputs and check the analysis. These tasks do not directly produce outputs for the final report. Especially for students with limited experience in cost–benefit analysis or Python, we strongly recommend completing them before interpreting the mandatory results.
Mandatory Task Specifying core activities and deliverables that every group is expected to complete and include in the final report.
Advanced Task (optional) Suggesting extensions for groups interested in PRIM scenario discovery of economic and requirement failure mechanisms (Sections 4.3–4.4) or in searching the monitoring thresholds of the adaptive plan (Section 5.1).

Use the same paired scenario identifiers for all plan comparisons. Report monetary results together with their price basis, discount rate, time horizon and units. Keep performance requirements, unpriced effects and implementation commitments visible rather than compressing them into one score. The MehrSpur values quoted in this sheet are the supplied defaults. You can replace them with the values of your own case study.

1 Part 1: Define and prepare the appraisal

1.1 Planning requirements and analysis handover

Notebook 05 should use the saved Notebook 04 experiment whenever it is available. This preserves the same sampled futures, uncertainty trajectory shapes, monitoring rules, performance requirements and plan-specific implementation dates. The handover checks the saved results before accepting the comparison.

For the MehrSpur example there are two requirements. The public-transport share of passenger-kilometres must exceed 35% from Year 20 onwards. The peak load/capacity ratio on the PT counting section must stay below 110% in every year of the horizon. This requirement exists only when the counting section and PT crowding are switched on. Economic value and objective protection remain separate.

Mandatory Task

Confirm a valid handover from Notebook 04.

- Record the scenario sample, seed, uncertainty trajectories, requirements, trigger rules and implementation dates used in the appraisal.
- Verify that the handover checks pass or explain every discrepancy before interpreting plan comparisons.
- State the assessment years, units and preferred direction for every active requirement.
- Explain where monetised impacts overlap conceptually with planning requirements and how double counting is avoided.

1.2 Select plans and display indicators

Select the plans to display, the adaptive plan and its predetermined comparators. In the MehrSpur example the comparators are Staged 1 and Static 2 (COMPARISON_PLANS). They are reused in Parts 3 and 4 and in Section 5.1. All evaluated plans, including the Baseline, remain in the regret benchmark even when only a subset is shown in a figure. The indicator catalogue defines labels, units and favourable directions for monetary and physical outcomes; edit the directions to match your own planning objectives.

Supporting Task (optional)

Check that the displayed comparison is internally consistent.

- Describe the baseline, static, predetermined staged and adaptive plans in the same terms used in Notebook 04.
- Verify the units and favourable direction of each selected display indicator.
- Use the same comparator definitions throughout the value-of-waiting analysis.

1.3 Build or load the paired appraisal dataset

1.3.1 Scenario sampling and paired comparisons

Every plan is evaluated under the same sampled future. With the Notebook 04 handover enabled, saved Latin Hypercube scenarios and their identifiers are reused. In standalone mode, scenario uncertainties are selected from the registered parameters, transient inputs follow the chosen trajectories and non-transient inputs retain their sampled values. Appraisal-sensitivity parameters are varied separately in Section 4.5.

1.3.2 Matched welfare and the Rule of a Half

Matched welfare means that travel quantities and time components are kept separately for each origin–destination pair and mode before aggregation, so that each change is valued where it occurs. Travel-

time benefits use person-trips and the Rule of a Half: existing users gain the full change in generalised travel cost, while users who switch or are newly attracted are credited, on average, with half of it. Car, cycling and walking time are included together with PT in-vehicle, waiting, access, egress and transfer-walking time. Road distance-based externalities use vehicle-kilometres (car passenger-kilometres divided by occupancy); train externalities follow the configured train service rather than passenger numbers.

1.3.3 Counting section / external flow and annualisation

The optional PT counting section identifies modelled journeys using the section. An external flow can be added to fill the difference between that count and the observed total. This external flow, called supplementary passengers in the notebook outputs, is calibrated once in Notebook 02 against nominal baseline counts (120 000 daily passengers for MehrSpur). Supplementary passengers receive the full section time savings, while modelled users keep their matched origin–destination benefits. Ordinary travel time is converted from the modelled peak period to a day with mode-specific factors and to a year with 300 equivalent days. Road congestion and PT crowding accrue only during the peak hours (nominal 4 per day; 3 and 5 are tested in Section 4.5). Crowding adds a penalty to the affected section minutes that grows with load above capacity: the time is weighted by $M = \min(3, 1 + 2 \times \max(Q/C - 1, 0))$, so it is at most tripled.

1.3.4 Option premiums, residual value and construction emissions

Adaptive option premiums are paid at time zero, even when a trigger never fires. Ordinary CAPEX is paid when a package opens, and operating and maintenance costs accrue while it operates. A package whose opening falls after Year 40 therefore carries no CAPEX inside the appraisal, although its in-horizon construction emissions are counted; such commitments are reported separately in Part 3. The final appraisal year credits a residual value for the remaining life of eligible capital (MehrSpur example: 60% of each package's capital with an 80-year life, ageing from opening), discounted from the end of the horizon. Construction emissions are spread evenly over the construction lead time, valued with the carbon price of each year and reported separately from CAPEX and operating emissions; an untriggered package emits nothing.

1.3.5 Appraisal assumptions and coverage

Document the discount rate, price basis, annualisation factors, values of time, carbon values, accident and air-pollution values, CAPEX and OPEX assumptions, asset lifetimes, residual-value shares and adaptive premiums. The supplied appraisal does not fully represent vehicle operating resources, additional train resources, reliability, construction disruption, ecology, distributional effects or every health impact. Omitting an effect from the monetary model does not make its value zero; record it in the unpriced-effects register (Section 5.2).

Mandatory Task

Prepare a reproducible paired life-cycle appraisal.

- Use identical scenarios for every plan and confirm that scenario IDs and implementation histories remain aligned.
- Document the welfare calculation, cohorts, modes, time components, annualisation factors and occupancy assumptions.
- State the timing of CAPEX, OPEX, option premiums, construction emissions and residual-value credits.
- Record the monetary price basis, discount rate, appraisal horizon, asset lives and all case-study adaptations.

1.4 Does the approximation preserve appraisal benefits?

Notebook 05 uses the Gaussian-process (GP) surrogate from Notebook 03 instead of running the full traffic assignment for every scenario. The saved GP-versus-assignment checks compare annual matched baseline/project benefits and selected physical outcomes. They do not validate every scenario NPV or remove the need to review interpolation and accounting assumptions. The supplied acceptance rule allows a mean absolute total-benefit error of at most 2% of the mean absolute reference benefit. The transport, assignment-convergence, large-error and benefit-sign checks must also pass. Component-level errors remain diagnostic.

Mandatory Task

Establish whether the approximation is adequate for appraisal.

- Report the total-benefit accuracy metric and the stated acceptance threshold.
- Review transport, assignment-convergence, benefit-sign and large-error diagnostics together.
- Investigate large errors in small components without treating percentage error alone as decisive.
- Explain the remaining limitation: the checks are sample-based annual comparisons, not a guaranteed error bound on every NPV.

2 Part 2: Compare economic value, risk and protection

2.1 NPV distributions across plans

Incremental NPV compares each plan with the Baseline in the same scenario. The empirical cumulative distribution function (ECDF) shows the full sampled distribution and the probability of falling below the zero-NPV (break-even) boundary. The boxplot summarises medians, interquartile ranges and downside tails. A first-order dominance claim requires one empirical curve to remain to the right without crossing.

Mandatory Task

Compare the distribution of incremental economic value.

- Report mean and median NPV, the share of scenarios with positive NPV (break-even share) and the lower percentiles (P10, P25) for every plan.
- Identify curve crossings and avoid claiming dominance when performance depends on the scenario region.
- Relate probabilities to the selected uncertainty distributions and sample; do not present them as universal frequencies.

2.2 NPV component breakdown

Decompose incremental NPV into user and societal benefits, construction climate cost, direct project costs and residual value. Benefits include the configured mode-specific time components, crowding relief and changes in climate, air pollution, noise and accident externalities. Direct costs include package CAPEX, operation and maintenance and adaptive option premiums. The displayed benefit–cost ratio is the ratio of ensemble mean benefits to ensemble mean net agency costs, not the mean of scenario-specific ratios.

Each component is measured as the Baseline's discounted cost minus the plan's discounted cost ($NPV_k = NPC_{k,\text{baseline}} - NPC_{k,\text{plan}}$, where NPC is net present cost). Positive values are benefits of the plan; negative values are costs.

Mandatory Task

Explain why the plans create or destroy economic value.

- Report the main positive and negative NPV components in MCHF and retain the complete exported component table.
- Keep construction carbon, operating emissions, CAPEX, OPEX, option premiums and residual value distinct.
- Interpret the benefit–cost ratio.

2.3 Component-level NPV ECDF

Component-level empirical distributions reveal uncertainty that is hidden by mean stacked bars. Two plans can have similar mean travel-time benefits yet very different lower tails, while a small average component can still matter in particular scenarios.

Supporting Task (optional)

Inspect the distributions of important NPV components.

- Compare their sign, spread and tail behaviour across plans.
- Identify which components explain crossings or downside risk in the total NPV distributions.
- Use the sign convention of Section 2.2 throughout: positive values favour the plan.

2.4 Parallel-coordinates multi-dimensional performance

Parallel coordinates compare economic value and requirement-related indicators across the same scenarios. All displayed axes place favourable outcomes at the top, while exported values retain physical units. Median profiles show central performance; they do not establish robustness or describe one coherent realised future.

Supporting Task (optional)

Use physical and monetary indicators together without collapsing them into one score.

- Select a small, interpretable set of axes and verify their units and favourable directions.
- Compare median profiles with the downside, regret and requirement-attainment evidence in Section 2.5.

2.5 Economic value, regret and requirement robustness

For each scenario ω, economic regret is the NPV of the best tested plan in that scenario minus the NPV of the plan being assessed: $\text{Regret}(s, \omega) = \max_{s'} \text{NPV}(s', \omega) - \text{NPV}(s, \omega)$. The Baseline (NPV = 0) is one of the tested plans, so a plan with negative NPV always has a regret at least as large as its loss. Downside loss is the amount by which NPV falls below zero; the mean loss below zero averages $\max(-\text{NPV}, 0)$ over all scenarios, counting profitable scenarios as zero. Tail means average the worst 10% of scenarios for each plan. Regret measures forgone opportunity; downside measures failure to beat the Baseline. Both depend on the tested plan set and the sampled scenarios. Requirement attainment and failure duration provide a separate dimension of robustness.

The joint comparison distinguishes scenarios with positive NPV and protected requirements, economic success only, requirement protection only, and neither. A plan is non-dominated in a displayed mean-NPV/requirement-protection screen when no other plan is at least as good on both and strictly better on one.

Mandatory Task

Assess economic and performance robustness together.

- Report the mean loss below zero, the mean NPV of the worst 10% of scenarios, mean regret and maximum sampled regret.
- Report attainment and failure duration for each active requirement, not only an all-requirements aggregate.
- Present the four joint economic/requirement outcome classes for the main plans.
- Explain how the baseline, tested plan set, sample and adaptive premium affect the regret and robustness results.

3 Part 3: Understand the value and price of waiting

Adaptive implementation can defer or avoid expenditure, but it can also defer or forgo transport benefits and requirement protection. Every comparison in this part is paired within the same scenario; in the MehrSpur example the Adaptive plan is compared with Staged 1 and Static 2. Positive component differences favour the Adaptive plan. A commitment whose opening lies beyond Year 40 carries no in-horizon CAPEX and is distinct from a package that is never triggered.

3.1 Savings, economic value and objective protection

Compare expenditure savings with additional years in which at least one requirement fails, then test whether those same scenarios gain or lose NPV after forgone benefits are included. One calendar year counts once in the aggregate failure measure even if several requirements fail, so requirement-specific results remain necessary.

Mandatory Task

Identify when waiting is economical and when it weakens the plan.

- Use same-scenario comparisons with explicitly named predetermined comparators.
- Distinguish packages that open later, packages committed with an opening beyond the horizon and packages never committed, and use the component groups (CAPEX, OPEX, residual value, time and access, crowding, other effects) to show where savings and losses arise.
- Identify scenarios where expenditure is saved without weakening protection and scenarios where forgone benefits exceed the savings.
- Use the paired component table to explain the source of each important trade-off.

3.2 Downside protection and investments avoided

Focus on scenarios in which the predetermined comparator loses money and determine whether adaptation improves NPV by avoiding, delaying or resizing commitment. Avoiding a package does not guarantee positive NPV because the flexibility premium is paid in every scenario.

Mandatory Task

Evaluate downside protection without hindsight bias.

- Report the full ensemble and the adverse-comparator subset.
- Distinguish packages opened, committed beyond the horizon and never committed.
- Explain paired NPV gains or losses using component differences measured from zero.
- State clearly that realised economic outcomes are diagnostic and were not available to the monitoring rule.

4 Part 4: Explain results and test assumptions

4.1 Directional driver check

Spearman rank correlation provides a first directional check of how sampled scenario inputs relate to incremental NPV and its components. A positive value means that higher sampled input values are associated with a higher contribution. It is descriptive, can be confounded by correlated responses and is undefined for constant outcomes.

Mandatory Task

Identify and interpret the main directional economic drivers.

- Report the largest positive and negative rank correlations for the selected plans and outcomes.
- Connect the signs to the relevant welfare, cost or externality components.

4.2 Economic and requirement vulnerability

Economic vulnerability is defined as NPV below zero relative to the same scenario baseline. Performance vulnerability is a breach of an active requirement during its assessment years. These mechanisms can overlap but are not interchangeable: a plan may be profitable while failing a strategic requirement or may protect the requirement while destroying economic value.

Mandatory Task

Keep economic and performance vulnerability mechanisms distinct.

- Report the frequency of negative NPV and each requirement failure for every selected plan.
- Explain the scenarios in the joint outcome classes using the component and driver evidence.
- Avoid treating requirement attainment as an additional monetary benefit.

4.3 PRIM scenario discovery by failure mechanism

The Patient Rule Induction Method (PRIM) searches for restricted regions (boxes) of the sampled uncertainty space in which negative NPV or a particular requirement failure is concentrated. Run each mechanism separately. PRIM runs only when each outcome class contains at least 15 scenarios. A box qualifies when at least 75% of the scenarios inside it fail (density) and it restricts at least one input; among qualifying boxes, the one covering the most failures is selected. If none qualifies, the unrestricted initial box is kept as a labelled reference.

Advanced Task (optional)

Use PRIM to investigate distinct vulnerability mechanisms.

- Select justified scenario inputs and report every restricted bound.
- Report box density, coverage and mass together with the overall failure frequency.
- Treat an unrestricted initial box, or a Baseline box shown for another plan (Section 4.4), as a reference, not a successful discovery.
- Do not interpret the absence of a qualifying box as proof that no vulnerability exists.

4.4 PRIM box comparison

Two-dimensional rectangles are projections of potentially higher-dimensional PRIM boxes. Density and coverage are calculated using all restricted dimensions, which must remain visible in the

accompanying table. When a plan has no qualifying box, the notebook may show a baseline or unrestricted reference with recalculated metrics; it remains a reference, not a discovery for that plan.

Advanced Task (optional)

Compare plan-specific boxes without hiding higher-dimensional restrictions.

- Explain similarities and differences in economic-loss and requirement-failure mechanisms.
- Report all bounds and indicate clearly when a rectangle is only a two-dimensional projection or a reference box taken from the Baseline.
- Discuss the limitations created by sample size, weak separation and uniform outcomes.

4.5 One-at-a-time appraisal-assumption sensitivity

Vary each registered sensitivity parameter separately at its P10 and P90 test values while all other inputs follow their nominal paths. Each plan is compared with a Baseline evaluated under the same assumption, which isolates the effect of appraisal assumptions on incremental NPV. For MehrSpur, for example, the number of peak hours is tested at 3 and 5 around the nominal 4. Some inputs are reference values rather than annual values: carbon is set for a reference year and then escalated by calendar year, so a constant reference input does not mean a constant annual carbon price. The analysis does not represent interactions or the joint probability of assumptions. The tornado plot reports the selected plan's change from its nominal NPV.

Mandatory Task

Test whether plausible appraisal assumptions overturn the conclusion.

- Document every tested parameter, unit, reference basis, nominal value and low/high case.
- Report the NPV change and whether plan ranking, break-even status or the recommendation changes.
- For each parameter, state whether the tested value is a constant input, a reference-year value that is escalated, or a transient path.
- State the limitation of one-at-a-time analysis and identify interactions that merit additional testing.

5 Part 5: Improve the plan and recommend

5.1 Explore monitoring thresholds

The optional threshold search replays one adaptive plan under the same prepared scenarios while changing one or two monitoring thresholds. Indicators, persistence rules and construction lead times remain fixed. The best tested thresholds maximise mean NPV within the sample, after which requirement protection, downside, regret and commitments must be checked. A boundary result suggests that the search range should be extended; an in-sample optimum is not evidence of performance on new scenarios.

Regret in this comparison is computed over all existing plans plus the best tested policy, so it can differ from the values in Section 2.5.

Advanced Task (optional)

Explore whether monitoring thresholds can improve the plan.

- Verify that the current-threshold replay matches the saved appraisal before searching.
- Report search ranges, resolution, best tested thresholds and whether the optimum lies on a boundary.
- Compare mean NPV, requirement protection, downside, regret and package commitments against the original policy.
- Treat the in-sample optimum as a candidate only, and state that it needs testing on new scenarios (for example, another seed) before it is recommended.

5.2 Interpret the appraisal results

The notebook's summary table reports mean NPV, the share of scenarios with positive NPV, requirement protection and their joint attainment for each plan. Record your interpretation for each plan in INTERPRETATION_NOTES. Meeting a requirement is not an additional monetary benefit. Conflicts between economic value, downside, regret and requirement protection must be made explicit rather than resolved by an undocumented score. Chapter 8 describes how to combine this interpretation with Notebooks 02–04 into the final case.

Mandatory Task

Prepare an evidence-based recommendation.

- State the recommended plan and the baseline and comparator plans against which it was judged.
- Support the recommendation with central NPV, downside, regret, component and requirement-protection evidence.
- Explain when adaptation creates value, when it delays benefits and which commitments remain beyond the horizon.
- Identify the scenario drivers and appraisal assumptions that could overturn the recommendation.
- Specify any further validation, data collection or policy adjustment needed before implementation.

6 Key outputs for Notebook 5

Mandatory Task

By the end of Phase 5, every group should retain:

- a documented Notebook 04 handover with aligned scenarios, requirements, trigger rules and implementation histories;
- a reproducible register of appraisal assumptions, units, price basis, discount rate, horizon and unpriced effects;
- the paired appraisal dataset and the compact approximation-validity checks;
- NPV distributions, the share of scenarios with positive NPV and lower-percentile summaries for all compared plans;
- the complete NPV component breakdown, including option premiums, construction carbon and residual value;
- economic regret, requirement attainment, failure duration and joint economic/performance outcome tables;
- paired value-of-waiting results that distinguish deferral, avoided implementation and beyond-horizon commitments;
- directional driver and vulnerability-mechanism evidence, with PRIM outputs if undertaken;
- one-at-a-time appraisal sensitivity results and any identified decision reversals;
- the tested threshold-policy comparison if the advanced improvement task is undertaken; and
- a concise, evidence-based recommendation with limitations and follow-up actions.

Completed figures and tables are exported to figures/05_appraisal/ and results/05_appraisal/. Rerunning an analysis replaces matching exports, so retain the configuration and code version used for the final report.

7 Connection to SIA

Notebook 5 supports strategic infrastructure appraisal by connecting adaptive planning with transparent societal valuation under deep uncertainty. It extends a conventional single-forecast cost–benefit calculation into a paired comparison of economic value, objective protection and implementation flexibility across plausible futures.

In particular, the work supports:

- documenting price bases, discounting, annualisation and valuation assumptions;
- separating user benefits, externalities (valued with NIBA-based coefficients, the Swiss sustainability indicators for rail infrastructure projects), construction climate effects and agency costs;
- testing whether an investment remains worthwhile across uncertain demand, behaviour, technology and cost conditions;
- comparing downside loss and regret rather than relying only on expected NPV;
- keeping strategic requirements and unpriced effects visible beside the monetary appraisal;
- evaluating whether flexibility avoids lock-in or merely postpones beneficial action;
- identifying vulnerability mechanisms and assumptions that warrant monitoring or further evidence; and
- providing a traceable basis for selecting, adapting or rejecting an infrastructure plan.

The recommendation must still be interpreted with stakeholder knowledge, legal and institutional constraints, financing, distributional effects, constructability and impacts outside the model boundary.

A positive NPV is evidence of estimated societal value under the documented assumptions; it is not, by itself, a complete decision rule.

8 Bringing the analyses together: from Notebook 2 to Notebook 5

The basis for the final analysis was progressively established in the earlier phases: the system boundary, planning objectives, performance indicators, infrastructure interventions, candidate plans, uncertainties, performance requirements, monitoring indicators, implementation constraints and appraisal assumptions. These choices determine which questions the analysis can answer. They should therefore be stated clearly and related to the original planning problem. A convincing final report should begin by explaining what decision must be made, which alternatives are being compared, what successful performance means and which aspects of the future remain uncertain.

(Notebook 02, Parts 1-4; Notebook 04, Sections 1.1-1.3; Notebook 05, Sections 1.1-1.3.)

The simulations provide the evidence for the final interpretation. Rather than evaluating the alternatives under one forecast, the analyses explore an ensemble of plausible futures. Across these futures, uncertain conditions develop differently, infrastructure measures may be implemented at different times, and the plans produce different combinations of system performance, requirement attainment, expenditure and societal value. The ensemble is not intended to predict which exact future will occur. Its purpose is to reveal recurring behaviours: under which conditions a plan performs well, when it becomes vulnerable, which uncertainties matter most and whether implementation or adaptation occurs early enough.

(Notebook 03, Parts 1-7; Notebook 04, Parts 2-5; Notebook 05, Parts 2-4.)

The final report should use this evidence to select and explain a number of representative futures/pathways. These should not be invented anecdotes or descriptions of isolated, convenient model runs. Each should represent a recognizable type of future found in the simulation results and connect the uncertain conditions to system behaviour, plan implementation and decision consequences.

For example, one representative future might describe sustained demand growth and increasing capacity pressure, resulting in early intervention and substantial user benefits. Another might describe moderate growth or technological change, under which a major fixed investment becomes unnecessary or economically unattractive. A third might reveal a difficult future in which requirements are breached despite intervention because implementation is too slow or because the selected intervention does not address the dominant failure mechanism. The relevant futures and mechanisms will depend on each group's case study.

Below, we present a set of questions that might help you prepare the presentation and the report. These are guiding questions meant to help build a story about the analysis you've conducted. Not all of these questions might be relevant for your own case study, and there might be some that are missing!

- **What planning problem is being addressed, and what is its current state?** Define the system and its users, describe the baseline conditions and main deficiencies, and state the objectives, indicators and performance requirements that motivate action.
  - *(Notebook 02, Sections 1.1-1.7; Notebook 04, Sections 1.1-1.2.)*
- **Which intervention types and alternatives are being considered, and what is each intended to achieve?** Explain the physical or operational mechanism of each intervention, how interventions are combined into stages or plans, and which system outcomes each alternative is expected to improve.
  - *(Notebook 02, Parts 3-4; Notebook 04, Sections 1.3-1.4)*
- **What combination of uncertain conditions characterizes the future?** Use the selected uncertainties, distributions, sampled scenarios and temporal trajectories.
  - *(Notebook 03, Parts 1-4; Notebook 04, Sections 1.5-1.7.)*

- **How do these conditions affect demand, modal shares, congestion, capacity or other relevant system outcomes?** Connect the uncertain inputs to the model behaviour and selected performance indicators.
  - *(Notebook 02, Sections 1.4-1.7 and 3.3; Notebook 03, Parts 5-7.)*
- **Which performance requirements are protected or violated, and when?** Report attainment, first-failure years, failure duration and any competing requirements.
  - *(Notebook 04, Sections 3.4 and 4.1-4.2; Notebook 05, Section 2.5.)*
- **How does each plan respond under the same future conditions and uncertainty trajectories, such as different growth patterns?**
  Use paired comparisons in which every plan is evaluated using the same sampled uncertainty values and trajectory shapes. This ensures that observed differences are attributable to the plans rather than to different scenario samples.
  - *(Notebook 03, Section 4.5 and Part 7; Notebook 04, Sections 1.5–1.7, 2.4–2.5 and Part 3; Notebook 05, Parts 2–3.)*
- **Which infrastructure measures are decided, constructed and opened in which cases?** Distinguish decisions from construction, operation, non-implementation and openings beyond the analysis horizon.
  - *(Notebook 04, Sections 5.1-5.2 and 5.5.)*
- **Does adaptation or staged implementation occur early enough to prevent or reduce failure?** Compare opening dates with first failures, restoration windows and construction-adjusted decision deadlines.
  - *(Notebook 04, Sections 4.4 and 5.4.)*
- **What are the main economic benefits, costs and unpriced effects?** Explain the NPV components, including user benefits, externalities, investment and operating costs, option costs, construction emissions and residual value. Record relevant effects that are not monetized.
  - *(Notebook 05, Sections 1.3, 2.1-2.2 and 5.2.)*
- **Which plan performs best, in which sense, and what trade-offs remain?** Consider mean and median NPV, downside risk, requirement protection, robustness, regret and the consequences of waiting. Do not select a preferred plan using only one indicator.
  - *(Notebook 05, Sections 2.5, 3.1-3.2 and 4.5.)*
- **What information should decision-makers monitor as the future develops?** Identify the observable indicators that provide useful warning and explain when new information could justify changing the plan or its timing.
  - *(Notebook 04, Parts 2 and 4, particularly Sections 2.0-2.2 and 4.4.)*

The recommendation should follow from the comparison of this evidence. For instance, a plan may have a high average NPV but perform poorly in a recognizable set of adverse futures. An adaptive plan may avoid unnecessary expenditure in some futures but respond too late in others. A predetermined or staged plan may provide earlier protection but create a risk of costly overinvestment. These differences should be discussed explicitly.

The objective is to identify the alternative that provides the most defensible balance of economic value, objective protection, implementation feasibility, flexibility and residual risk.

You could think about constructing an analytical line that follows this pattern:

Planning problem -> model representation -> uncertain futures -> system behaviour -> plan response and implementation -> physical and economic consequences -> comparison of alternatives -> recommendation.

As mentioned above, this is just meant to showcase an example of how you could present your analysis. Feel free to select what works best for you!

In any case, the final report should be read as one integrated analysis rather than a sequence of disconnected notebook summaries. It should explain not only what the model calculated, but also why the results matter, under which futures the conclusion holds, what could overturn it and what should be monitored before and during implementation.

A strong conclusion will state the recommended course of action while remaining transparent about uncertainty, unpriced effects, implementation constraints, model limitations and the conditions under which the recommendation should be reconsidered.